

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2]

1. If $SST = 90$ and $SSR = 70$, compute the coefficient of determination and interpret its Meaning.
2. Interpret the regression coefficient of regression model $y = 2 + 3x$. what is the estimated value when $x = 2$?
3. For a set of 8 data points, the sum of squared differences between observed values and estimated values of y is 35.72, what is the standard error of estimate?
4. Given information from a multiple regression analysis. if $n=25$, $b_1=5$, $b_2=10$, $sb_1=2$, $sb_2=8$. Setup a 90% confidence interval estimate of the population slope β_2 .
5. Using 2010 as the base year, compute price index number for 2015 by using unweight aggregates index method.

Product	A	B	C	D	E
2010 price (Rs)	27	32	290	60	21
2015 price (Rs)	52	51	314	76	86

6. What are the components of time series?
7. Demand for a certain commodity was 300 units in June, 60 in July and 160 in August. The forecast for June was 110 units with a smoothing constant of 0.30, what will be the September forecast.
8. What is the main objective of Assignment model? Differentiate between balanced and unbalanced assignment model.

9. Find the active and inactive constraints of following LP. Max profit $z=9X+7Y$.

Subject to,

$$2X+Y \leq 40$$
$$X+3Y \leq 30$$
$$X, Y \geq 0,$$



where optimal profit $z = 750$ at $X = 18$ and $Y = 4$.

10. What is the role of dummy activity in network model? Draw network diagram for the following project.

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Activities	A	B	C	D
Predecessors	-	-	A,B	B

Section "B"

Descriptive Answer Questions

Attempt **any six** questions. $[6 \times 10]$

11. The income and expenditure of 10 families is given below.

Expenditure \$	5	10	15	20	25	30	35	40	45
Income \$	20	25	30	30	30	40	50	50	70

- Fit the regression equation of expenditure on income, Estimate the expenditure of family having income \$65.
- Compute coefficient of correlation between income and expenditure and test the significance of correlation coefficient at 5% level of significance.

12. *“Fishers Index number is called an economic barometer”. Why? From the following data show that fisher’s ideal index number satisfies both the time reversal test and factor reversal test.*

Commodity	2011		2012	
	Price	Expenditure	Price	Expenditure
P	10	200	12	180
Q	12	144	15	300
R	8	80	8	96
S	20	300	12	432

13. The following table shows the population of a city for seven years.

Year	2016	2017	2018	2019	2020	2021	2022
Population(million)	23	24	28	31	33	35	38

- Find the trend line that best describes the data.
 - Find the monthly growth rate.
 - Calculate the trend values for different years.
 - Find the percentage of trend for the population of the city.
 - Find the relative cyclic residual for the population of the city.
 - In which year does the largest and smallest fluctuation from trend occur, and is it the same for both methods.
14. a) From the given information, calculate forecast for year 2015 using $\alpha=0.1$ and forecast for year 2012 was 95 units.

Year	Actual demand
2012	100
2013	120
2014	250

- b) A project work consists of four job for which 4 contractors have submitted tenders. The tender amounts quoted in lakhs of rupees are given in the matrix from.

Job/contractor	I	II	III	IV
A	10	16	12	9
B	24	22	20	26
B	30	28	32	34
D	15	12	10	16

Find the assignment which minimizes the total cost of the project. (Initial feasible solution by VAM)

15. Using the following data, find the minimum transportation cost.

Sources	Destinations				Availability
	A	B	C	D	
P	9	7	10	8	14
Q	8	11	9	11	27
R	13	10	12	10	14
Requirement.	15	19	11	10	55

16. A firm takes two types of furniture, chairs and tables. The contribution(profit) for each product as calculated by the accounting department is Rs. 200 per chair and Rs.300 per table. Both products are processed on three machines M1, M2 and M3. The time requirement in hours by each product and total time available in hours per week on each machine are as follows.

Machine	Chair	Table	Available time.
M1	3	3	36
M2	5	2	50
M3	2	6	60

How should the manufacturer schedule his production in order to maximize the contribution (profit)? Use graphical method.

17. A small project composed of eight activities and their time estimates are as follows: Consider the following project.

Activity	Pre-requisite Activity	Time Estimate		
		Optimistic	Mostly likely	Pessimistic
A	-	1	4	7
B	-	5	10	17
C	A	3	3	3
D	A	1	4	7
E	B	8	15	26
F	B	2	4	8
G	C,E	5	5	5
H	G,F	2	5	8

- a) Draw the network for this project and find the expected project completion time.



- b) What is the probability that the project will be completed within 41 days.
- c) What should be the scheduled completion time for the probability of completion to be 95%?

Case Analysis [20]

18. Successful selling is as much an art as a science, but many sales managers believe that personal attributes are important in predicting sales success. Design Alley is full service interior design store that sales custom blinds, carpets and wall coverings. The store manager, **Mr. A**, contracted with a sales- force selection company to conduct pre-hiring test on four aptitudes. **Mr. A** has collected sales growth data for 25 of the sales people who were hired, along with the scores from the four test of aptitude: **creativity, mechanical ability, abstract thinking, and mathematical calculation**. **Mr. A** generated the following Minitab output.

Coefficient Table

Predictor	Coefficient	Std. dev	t-ratio	p-value
Constant	70.0660	2.130	32.895	0.000
CREAT	0.42160	0.17192	2.4523	0.0024
MECH	0.27140	0.21840	1.2426	0.228
ABST	0.74504	0.28982	2.57	0.018
MATH	0.14955	0.06871	2.18	0.000

ANOVA Table

Source	Sum of Squares	df	Mean Square	F
Regression	1050.78	4	262.695	62.64
Residual	83.88	20	4.194	
Total	1134.66	24		

- Write the regression equation for sales growth in terms of the four factors test. Also predict the sales growth when CREAT = 12, MECH = 14, ABST = 18, and MATH = 30.
- Find standard error of estimate.
- How much variation in sales growth is explained by the aptitude tests?
- At a significance level of 0.05, which of the aptitude tests are significant explanatory variables for sales growth?
- Is the overall model significant as a whole?